

Facebook open sources Pythia for vision and language multi-modal AI models

Facebook's AI research division has announced the release of Pythia, a deep learning framework that supports multitasking in the vision and language domain, on GitHub.

This modular plug-and-play design enables data scientists to quickly build, reproduce and benchmark AI models. As Facebook's blog post states, Pythia – which is built on top of the company's open source PyTorch framework – is designed for vision and language tasks, such as answering questions related to visual data and automatically generating image captions.

It can show how previous state-of-art models achieved related benchmark results and quickly gauge the performance of new models.

In addition to multitasking, Pythia also supports distributed training and a variety of data sets, as well as custom losses, metrics, scheduling, and optimisers.



"Pythia smoothes the process of entering the growing sub-field of vision and language, freeing researchers to focus on faster prototyping and experimentation," Facebook's blog post stated.

Facebook plans to expand on Pythia's initial open source release by adding tools, tasks, data sets and reference models.

While Rick O'Connor is the founder and CEO of the OpenHW Group, Rob Oshana, VP, software engineering, NXP, will serve as the chairman of the group's board.

Keysight, Nokia aim to accelerate development of automation solutions with OpenTAP

Keysight Technologies Inc. and Nokia have jointly launched an open source test automation project for accelerating software development. OpenTAP (the Open Test Automation Project) represents Keysight's first foray into stewarding open source software. It provides an open source, scalable architecture that enhances and accelerates the development of automation solutions within the test and measurement (T&M) ecosystem.



OpenTAP includes a growing community of developers dedicated to the idea of effortless automation. The core technology is a test sequencer, to which the community is continuously adding plugins and solutions that lead to faster development, new functionality and optimisation when testing.

Through these contributions and the enhanced collaboration possible with its open-core approach, OpenTAP aims to inspire creators to come up with innovative automation solutions in T&M and beyond.

"Our goal with OpenTAP is to enable unprecedented levels of collaboration in the test and measurement industry, and to benefit all participants in this open source project," said Jay Alexander, chief technology officer of Keysight Technologies.

Erja Sankari, VP of Nokia's supply chain engineering, added, "For many years, Nokia has relied on either closed, commercial test-automation products, or completely home-grown platforms that consume precious resources, in terms of budgets, support and maintenance. Building on OpenTAP fosters innovation in test optimisation and efficiency, ultimately reducing our cost-of-test and project risks."

OpenTAP originated from Keysight's internal project called Inner Source, which the company started to increase software reuse within its engineer groups.

"Coinciding with Keysight's Inner Source adoption, multiple customers (led by Nokia) approached Keysight to release core components of our test automation solutions as open source, in particular to enable more sophisticated testing in-place, in deployed devices. This confluence of capabilities and requirements led to the launch of OpenTAP," the OpenTAP team wrote in the official website.

The project is inspired by 'gypsy brewers' who collaborate with established breweries to create their own recipes. "For over a decade the tenets behind our work have always been focused on simplicity, scalability and speed for users. With OpenTAP, we open the doors to brewing something great together," the team wrote.

OpenTAP is released under the OEM-friendly Mozilla Public License Version 2.0. Keysight and Nokia previously demonstrated early capabilities and outcomes of OpenTAP at MWC19 Barcelona, in February 2019.